DO NOT ANSWER ON THIS SHEET

Stony Brook Institute at Anhui University

ISE 218: Fundamentals of Information Technology – Spring 2025

Quiz 2 – Unit 3

Time: 50 minutes

Neatly write your answers in English. For mathematical calculations you must show all work to receive full credit. You need to simplify your answers as much as possible, giving a single number as your answer.

Question 1 (6 pts) Write 3 MARIE instructions for Load, Store, and Add. You need to write both the assembly language instructions and binary bit patterns. Include your choice of addresses as well.

```
LOAD 0x100: 0001 0001 0000 0000

STORE 0x101: 0010 0001 0000 0001

ADD 0x102: 0011 0001 0000 0010
```

Other addresses are fine as long as the address and the bit pattern match.

Question 2 (4 pts) What is the purpose of the Program Counter (PC) in a CPU? Explain your answer.

The Program Counter (PC) holds the address of the next instruction to be executed.

Question 3 (3 pts) Write the RTL for the instruction `Subt X`.

```
MAR <- X

MBR <- M[MAR]

AC <- AC - MBR
```

Question 4 (6 pts) What is the difference among MAR, MBR, and AC in Marie Architecture?

Register	Full Name	Purpose
MAR	Memory Address Register	A 12-bit register that holds the memory address of an instruction or the operand of an instruction.
MBR	Memory Buffer Register	A 16-bit register that holds the data after its retrieval from, or before its placement, in memory.
AC	Accumulator	A 16-bit register that holds a conditional operator (e.g., “less than”) or one operand of a two-operand instruction (e.g., “add”).

Question 5 (6 pts) Suppose that a 2M × 16 main memory is built using 256K × 8 RAM chips and memory is word addressable.

a) How many RAM chips are necessary?

Total memory = 2M × 16 bits = $2 \times 2^{20} \times 16$ bits

Each RAM chip = 256K × 8 bits = $2^{18} \times 8$ bits

To get 16-bit words → need 2 chips per word (for 8 bits × 2 = 16 bits)

Number of words = 2M = $2 \times 2^{20} = 2^{21}$ words

So:

Total RAM needed in bits = $2^{21} \times 16 = 2^{21} \times 2^3 \times 2^1 = 2^{25}$ bits

RAM per chip = $2^{18} \times 8 = 2^{21}$ bits

Number of RAM chips = $2^{25} / 2^{21} = 2^4 = 16$ chips

b) How many address bits are needed to address one byte in a RAM chip?

Each chip has 2^{18} bytes (since it's 256K × 8 bits = 256K bytes)., so we need 18 bits.

c) How many address bits are needed for all memory?

Memory is word addressable and has 2M words = $2 \times 2^{20} = 2^{21}$ words, so we need 21 bits.

Instruction Number		Instruction	Meaning
Bin	Hex		
0001	1	Load X	Load the contents of address X into AC.
0010	2	Store X	Store the contents of AC at address X.
0011	3	Add X	Add the contents of address X to AC and store the result in AC.
0100	4	Subt X	Subtract the contents of address X from AC and store the result in AC.
0101	5	Input	Input a value from the keyboard into AC.
0110	6	Output	Output the value in AC to the display.
0111	7	Halt	Terminate the program.
1000	8	Skipcond	Skip the next instruction on condition.
1001	9	Jump X	Load the value of X into PC.